

CHAPTER 4:

Adaptations and Behaviour

(The extraordinary strategies of marine life)



Chapter	Big Question	Concepts	Examples	Visuals	Leads Into
Introduction	What counts as an adaptation?	Evolution	Extreme species	Before/after adaptations	Core
Adaptation Basics	Why do adaptations evolve?	Natural selection	Various species	Evolution diagram	Physical adaptations
Camouflage	How do species hide?	Mimicry	Octopus	Camouflage examples	Behaviour
Colour Change	Why change colour?	Communication	Cephalopods	Sequential images	Behaviour

Bioluminescence	Why produce light?	Deep sea adaptation	Lanternfish	Deep sea diagrams	Communication
Regeneration	Can organisms regrow body parts?	Regrowth	Sea cucumber	Regeneration cycle	Survival
Communication	How do marine species communicate?	Sound, signals	Whales	Communication diagrams	Social behaviour
Social Structures	Why form groups?	Schooling	Fish schools	Behaviour images	Migration
Migration	Why travel huge distances?	Navigation	Turtles	Migration maps	Navigation
Navigation	How do species find their way?	Magnetism	Turtles	Earth magnetic field	Threats

This chapter explores how marine organisms have evolved extraordinary physical, physiological and behavioural strategies to survive, feed, avoid predators, reproduce and find their way across the ocean. From the ability to vanish into the seafloor to the ability to navigate thousands of kilometres using invisible magnetic fields, adaptation and behaviour explain much of what makes ocean life so remarkable.

By the end of this chapter, you will have a better understanding of:

- What an adaptation is, and how natural selection produces it over generations.
- How camouflage and rapid colour change help marine animals hide, hunt and communicate.
- Why bioluminescence is so common in the deep ocean, and what it is used for.
- How some marine animals can regrow entire organs after injury.
- How whales, fish and turtles communicate, group together, migrate and navigate across entire ocean basins.

Introduction:

What counts as an adaptation?

The word "adaptation" is often used loosely in everyday language: a person might say they "adapted" to a new school or a new job. In biology, however, adaptation has a precise meaning. An adaptation is an inherited trait, shaped over many generations by natural selection, that improves an organism's chances of surviving and reproducing in its particular environment. Adaptations are not chosen or learned in a single lifetime; they accumulate slowly, as individuals with useful traits survive and pass those traits on to their offspring.

Marine environments range from sunlit coral reefs to the crushing darkness of ocean trenches, and organisms living in each of these settings show adaptations tailored to their surroundings. Adaptations are usually grouped into three broad categories:

- Structural (physical) adaptations: body parts or features, such as a streamlined body, gills, or a hard shell.
- Physiological adaptations: internal processes, such as producing antifreeze proteins or tolerating low oxygen.
- Behavioural adaptations: actions an animal performs, such as migrating, schooling or hunting in a particular way.

India spotlight

In April 2024, scientists from India's National Institute of Ocean Technology (NIOT) and the National Centre for Polar and Ocean Research (NCPOR) captured the first-ever image of an active hydrothermal vent in the Indian Ocean, at a depth of about 4,500 metres on the Central Indian Ridge.

The discovery is part of India's Deep Ocean Mission, which uses an Autonomous Underwater Vehicle (AUV) launched from the research vessel Sagar Nidhi to search a stretch of underwater mountain ranges scientists describe as "as rugged as the Himalayas."

Some of the most striking examples of adaptation come from extreme marine habitats. At deep-sea hydrothermal vents, water rich in toxic hydrogen sulphide and, in some cases, arsenic pours out of the seafloor at temperatures far too hot for most life. Giant tube worms such as *Riftia pachyptila* have no mouth, stomach or digestive tract at all: instead, they house colonies of chemosynthetic bacteria inside a specialised internal organ called a trophosome, which convert the vent chemicals into usable energy. In the western Pacific, the vent worm *Paralvinella hessleri* has evolved a way to combine the arsenic and sulphide in the water into a far less toxic mineral inside its own cells, allowing it to live in the hottest part of the vent that no other animal can tolerate.

Species Spotlight

Giant Tube Worm (*Riftia pachyptila*): The Fastest-Growing Animal Without a Mouth

Giant tube worms can grow more than 30 inches (about 85 centimetres) in a single year, making them among the fastest-growing invertebrates ever recorded. They anchor themselves permanently to the seafloor near hydrothermal vents, at depths of roughly 2,000 to 3,000 metres, and can live for decades in total darkness.

Source: BOEM: Tube Worms

Pompeii Worm (*Alvinella pompejana*): The Most Heat-Tolerant Animal Known

The Pompeii worm lives directly on the walls of hydrothermal vent chimneys, where its tail end can reach temperatures far hotter than any other animal is known to tolerate, while its feathery head pokes into cooler water just centimetres away. It survives partly thanks to a coat of bacteria living on its back, in a relationship scientists are still working to fully understand.

Source: *Science (AAAS): Deep-sea worms fight poison with poison to survive in hydrothermal vents*



FIGURE 1: GIANT TUBE WORM (RIFTIA PACHYPTILA)

These extreme adaptations illustrate an important idea that will run through this chapter: an adaptation only makes sense in the context of the specific environmental pressures an organism faces. A trait that is essential for survival at a hydrothermal vent would be useless, or even harmful, on a coral reef, and vice versa. As you move through the sections that follow, look for this same pattern: a challenge in the environment, followed by a trait that has evolved in response to it.

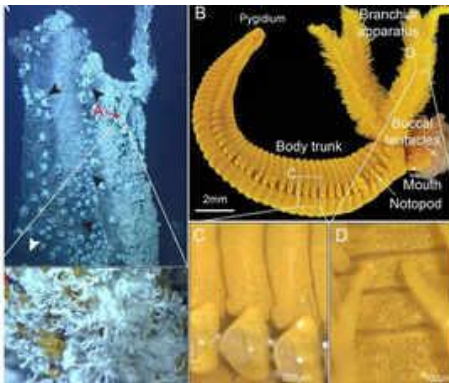


FIGURE 2: THE POMPEII WORM

Key Terms

Adaptation: An inherited trait that improves an organism's chance of survival and reproduction in its environment

Structural adaptation: A physical body feature shaped by evolution, such as fins or shells

Physiological adaptation: An internal biological process that helps an organism survive, such as producing antifreeze compounds

Behavioural adaptation: An inherited pattern of behaviour that improves survival, such as migration or schooling

Chemosynthesis: The process by which some bacteria produce energy from chemicals rather than sunlight

Extremophile: An organism adapted to survive in conditions that would be lethal to most other life

Things to Remember

1. An adaptation is an inherited trait shaped by natural selection over many generations, not something learned in one lifetime.
2. Adaptations are usually structural, physiological or behavioural.
3. Extreme marine habitats, such as hydrothermal vents, show some of the most dramatic adaptations in nature.
4. An adaptation only makes sense in relation to the specific challenges of an organism's environment.

Quick Quiz

1. What is the biological definition of an adaptation, and how is it different from the everyday use of the word?
2. Name the three broad categories of adaptation and give a marine example of each.
3. Why can giant tube worms survive without a digestive system?
4. Why might a hydrothermal vent adaptation be useless on a coral reef?

SECTION 1

Adaptation Basics

Why do adaptations evolve?

Adaptations evolve through a process called natural selection, first described by Charles Darwin. Natural selection works through three simple ingredients: variation (individuals in a population are not identical), heredity (some of that variation can be passed on to offspring), and differential survival and reproduction (individuals whose traits suit their environment tend to survive and reproduce more successfully than others). Over many generations, this means that helpful traits become steadily more common in a population, while less helpful ones become rarer.

India Spotlight

Mudskippers of the Sundarbans

Mudskippers are amphibious fish that spend most of their time out of water on the mudflats of mangrove ecosystems; of the ten genera found worldwide, nine occur in India, along both the east and west coasts.

They breathe partly through their skin and the lining of their mouth as long as these stay moist, and use specially adapted pectoral fins like small limbs to skip and crawl across mud and mangrove roots.

It is important to understand what natural selection is not. It is not a process that gives an organism what it "needs," and it does not act with foresight or intention. Instead, small, often random genetic differences between individuals turn out, purely by chance, to be more or less useful in a given environment. Marine environments apply constant selective pressure of this kind. Water is approximately 800 times denser than air, which makes streamlined bodies and efficient swimming mechanics far more important underwater than they would be on land. Pressure also increases rapidly with depth, roughly one additional atmosphere for every 10 metres, while temperature, salinity and light availability vary enormously between habitats, and

predators are everywhere. Species that happened to inherit traits suited to these pressures were more likely to survive long enough to reproduce.

Genetic research has confirmed this process in real marine populations. Studies of coastal species along the North American west coast, for example, have found strong genetic differences clustered around headlands and capes that mark the edges of a species' range: a signature of natural selection acting differently on populations exposed to different environmental conditions. Other studies have shown that even single changes in the amino acid sequence of a protein can adjust how that protein functions at different water temperatures, allowing closely related populations to specialise for warmer or cooler water.

Species Spotlight

Northern Elephant Seal: *A Body Built for Extreme Diving*

Northern elephant seals spend around 80 to 90 percent of their months-long foraging trips underwater, performing dive after dive with only brief pauses at the surface to breathe.

Before diving, they exhale most of the air from their lungs rather than holding a full breath, which helps prevent decompression sickness and reduces buoyancy so they sink more efficiently.

Source: Friends of the Elephant Seal: An Elephant Seal's Deep Dive

Antarctic Icefish (Family Channichthyidae): *The Only Vertebrates With Colourless Blood*

Icefish are the only known vertebrates that lack functional red blood cells and haemoglobin as adults, leaving their blood almost clear rather than red.

To survive without haemoglobin, icefish evolved unusually large hearts and expanded blood vessel networks, while antifreeze

glycoproteins bind to any ice crystals that enter their body and stop them from growing.

Source: California Academy of Sciences:
Antarctica's Antifreeze Fish



FIGURE 3: NORTHERN ELEPHANT SEAL

Key Terms

Natural selection: The process by which organisms with traits suited to their environment survive and reproduce more successfully, making those traits more common over generations

Variation: Differences between individuals of the same species

Heredity: The passing of traits from parents to offspring

Convergent evolution: When unrelated species independently evolve similar traits because they face similar environmental pressures

Things to Remember

1. Natural selection depends on variation, heredity and differential survival and reproduction: it does not plan ahead or act on purpose.
2. Marine environments exert strong, constant selective pressure through water density, pressure, temperature and predation.
3. Genetic studies have directly documented natural selection acting on real marine populations.

4. Unrelated species can independently evolve similar adaptations through convergent evolution when they face similar challenges.

Quick Quiz

1. What three ingredients does natural selection require to work?
2. Why is it inaccurate to say natural selection gives an organism what it "needs"?
3. Give one piece of genetic evidence that natural selection has shaped marine populations.
4. What physiological adaptations allow elephant seals to dive to great depths?

SECTION 2:

Camouflage

How do species hide?

Camouflage is a set of adaptations that allow an organism to avoid being detected, usually by blending into its background or by resembling something else entirely. In the ocean, where many predators hunt primarily by sight, camouflage is one of the most widespread survival strategies. Broadly, camouflage takes two forms: background matching, where an animal's colour and pattern resemble its surroundings, and mimicry, where an animal imitates the appearance or behaviour of another species, usually one that is dangerous or unpalatable to predators.

India Spotlight

Reef Stonefish of the Andaman Islands

The reef stonefish, found around the Andaman and Nicobar Islands, is considered the most venomous fish in the world, but it relies first on camouflage: its irregular, encrusted skin texture and mottled brown, orange or red colouring make it look like a lump of coral or an algae-covered rock. It rarely swims and will not flee when approached; instead it stays motionless on the seafloor as an ambush predator, only using its thirteen venomous dorsal spines as a last line of defence if stepped on or grabbed.



FIGURE 4: REEF STONE FISH

Cephalopods (octopuses, cuttlefish and squid) are among the most sophisticated masters of camouflage in the ocean. Their skin contains thousands of pigment-filled cells called chromatophores, each

surrounded by tiny muscles that can expand or contract the cell in milliseconds, changing the animal's colour almost instantly. Beneath the chromatophores, additional cell layers called iridophores and leucophores reflect light and add a white base layer, giving cephalopods an extraordinary range of colours, patterns and even textures.

Mimicry takes camouflage a step further than simple background matching: rather than blending into its surroundings, an animal imitates the appearance, colouring or even movement of a completely different species, usually one that predators tend to avoid because it is venomous, toxic or otherwise unpleasant to eat. Mimicry can involve changes to body shape as well as colour, and a small number of animals are known to switch rapidly between several different impersonations depending on which predator is nearby, rather than relying on a single fixed disguise. Related species sometimes share a similar strategy in different oceans: one species documented in the Caribbean has been found mimicking the colouring and swimming style of a flatfish, in what researchers described as the first such case recorded in the Atlantic.

Species Spotlight

Mimic Octopus (*Thaumoctopus mimicus*): *More Disguises Than Any Other Known Animal*

Divers have reported the mimic octopus impersonating far more than the three best-studied disguises, including anemones, jellyfish, feather stars, giant crabs, mantis shrimp and seahorses.

To imitate a lionfish, it spreads its arms wide and holds them rigid above its body, mimicking the venomous fanned spines of the real fish, and it is one of the only animals known to switch rapidly between several distinct impersonations rather than relying on a single disguise.

Source: *Ocean Conservancy: Why the Mimic Octopus Is the Ultimate Master of Disguise*

Tasselled Wobbegong Shark: *A Shark Disguised as the Seafloor*

Its name comes from an Australian Aboriginal word thought to mean "shaggy beard," a reference to the fringe of skin flaps around its jaw that break up its outline against the reef. Rather than chasing prey, it lies motionless for long periods and sometimes wiggles the tip of its tail to lure curious fish within striking distance, snapping shut with jaws that can dislocate to swallow prey nearly as large as itself.

Source: *Ocean Conservancy: Meet the Weird and Wonderful Wobbegong*



FIGURE 5: MIMIC OCTOPUS MIMICKING A FLOUNDER

Key Terms

Camouflage: An adaptation that helps an organism avoid detection by predators or prey

Background matching: A form of camouflage where an organism's appearance resembles its immediate surroundings

Mimicry: A form of camouflage or defence where one species imitates the appearance or behaviour of another

Chromatophore: A pigment-containing skin cell that cephalopods can expand or contract to change colour

Things to Remember

1. Camouflage generally works through background matching, mimicry, or a combination of both.
2. Cephalopods use chromatophores, iridophores and leucophores to change colour and pattern almost instantly.
3. The mimic octopus can impersonate several different dangerous species and switch between them depending on the threat.
4. Camouflage strategies are shaped by the specific predators and habitat an animal lives in.

Quick Quiz

1. What is the difference between background matching and mimicry?
2. How do chromatophores allow cephalopods to change colour so quickly?
3. Why does the mimic octopus switch between different disguises rather than using just one?
4. How does the tasselled wobbegong shark use both its appearance and its behaviour to catch prey?

SECTION 3:

Colour Change

Why change colour?

While Section 2 introduced colour change as a tool for camouflage, many marine species also change colour for reasons that have nothing to do with hiding. Rapid colour change is also used to communicate, to signal aggression, readiness to mate, submission, or alarm to other members of the same species, and cuttlefish are among the best-studied examples of this behaviour.

India Spotlight

Pharaoh Cuttlefish of the Arabian Sea

*The pharaoh cuttlefish (*Sepia pharaonis*) is India's most commercially important cuttlefish species, caught mainly off Kerala, Karnataka and Gujarat; catches in one west-coast state alone grew from around 3 tonnes in 1980 to nearly 9,000 tonnes by 2012.*

Like other cuttlefish, it uses muscle-controlled chromatophores to shift between a solid resting colour and a mottled brown-and-white pattern with a distinctive dark circle at its centre, a display used in both hunting and courtship.



FIGURE 6: PHAROAH CUTTLEFISH

A cuttlefish's skin contains millions of chromatophores, and researchers have found that these cells are controlled directly by a dedicated network of brain lobes that process visual

information and send precise signals down to the skin. Each chromatophore is attached to tiny radial muscles: when motor neurons fire, the muscles contract and stretch the pigment sac open, revealing colour; when the neurons stop firing, the sac relaxes back to its resting size. Because this signalling happens through direct nerve control rather than slower hormonal pathways, cuttlefish can shift an entire body pattern in a fraction of a second, and can display more than one function at once, for example camouflaging the body while flashing a contrasting display on one side to court a nearby mate while a rival male watches from the other side.

Researchers have also discovered that cuttlefish do not simply switch instantly to a perfect matching pattern. High-resolution imaging studies have shown that cuttlefish often try out several intermediate colour patterns in quick succession before settling on the one that best matches their surroundings, suggesting a more flexible, almost trial-and-error process rather than a single fixed reflex.

Species Spotlight

Peacock Flounder (*Bothus lunatus*): A Fish That Changes Colour by Watching, Not Just Feeling

This flatfish can match the colour and pattern of a new seafloor background in as little as two to eight seconds, one of the fastest documented colour changes in any fish, achieved through a combination of pigment cells and hormonal signals rather than the direct nerve control cephalopods use.

Its colour-matching ability depends entirely on sight: researchers found that a peacock flounder that cannot see its surroundings, because it is blind or has its eyes covered, is unable to camouflage itself and instead turns almost uniformly dark.

Source: Royal Society Open Science: Changeable camouflage: how well can flounder resemble the colour and spatial scale of substrates in their natural habitats?

Flamboyant Cuttlefish (*Metasepia pfefferi*):
Walking Instead of Swimming to Stay
Camouflaged

Unlike most cuttlefish, this small Indo-Pacific species often crawls along the seafloor using two of its lower arms, a slow, deliberate movement that keeps its outline from breaking the camouflage that fast swimming would disturb.

Its skin can flash bright red, purple, pink, yellow and white patterns, a display used both to mimic less appealing animals and to warn predators that its flesh is toxic.

Source: *Aquarium of the Pacific: Flamboyant Cuttlefish*

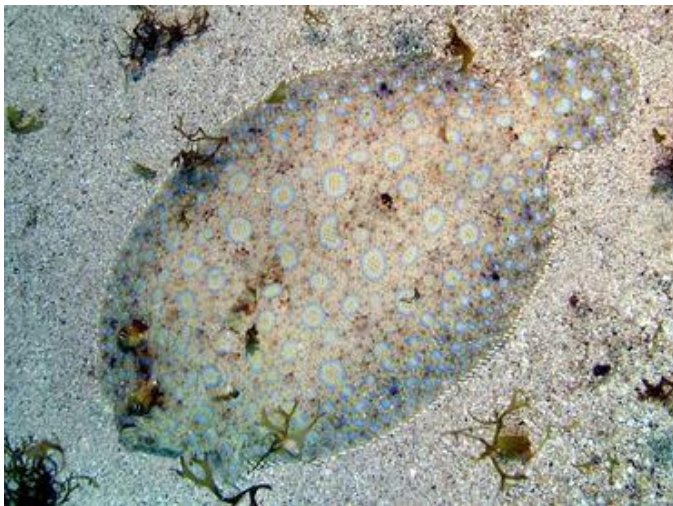


FIGURE 7: PEACOCK FLOUNDER

Key Terms

Iridophore: A skin cell that reflects light to produce shimmering or metallic colours

Leucophore: A skin cell that provides a white, light-scattering base layer beneath other pigment cells

Body pattern: A combination of colour, contrast, and texture that a cephalopod displays for camouflage or communication

Papillae: Small, muscular skin projections that some cephalopods use to change their skin texture

Things to Remember

1. Colour change in marine animals serves both camouflage and communication.
2. Cuttlefish chromatophores are controlled directly by the nervous system, allowing near-instant colour change.
3. Cuttlefish can display different patterns on different parts of the body at the same time.
4. Cuttlefish may try several intermediate patterns before settling on a final camouflage display.

Quick Quiz

1. Name two purposes of colour change in marine animals beyond camouflage.
2. How does nerve control allow cuttlefish to change colour faster than hormone-based systems could?
3. What role do iridophores and leucophores play alongside chromatophores?
4. How does the peacock flounder's colour-changing mechanism differ from a cuttlefish's, and what happens if it cannot see?

SECTION 4:

Bioluminescence

Why produce light?

Below about 200 metres, sunlight fades quickly, and by 1,000 metres the ocean is almost entirely dark. Yet this is far from an empty world: according to NOAA Ocean Exploration, roughly 80 percent of animals living between 200 and 1,000 metres depth are able to produce their own light, a phenomenon called bioluminescence. Because the deep sea is so vast, bioluminescence may be the single most common form of communication on the planet.

Bioluminescence works through a chemical reaction. An enzyme called luciferase acts on a light-emitting molecule called luciferin, and in the presence of oxygen this reaction produces light along with a harmless byproduct: with almost none of the energy lost as heat, unlike a typical light bulb. In some species the reaction happens inside the animal's own cells; in others, such as the Hawaiian bobtail squid, it is produced by colonies of bioluminescent bacteria that live inside a specialised light organ. Scientists estimate bioluminescence has evolved independently at least 40 separate times across different groups of marine life. In 2018, researchers found that ray-finned fishes alone evolved it at least 27 separate times, far more often than previously thought.

India Spotlight

Sea Sparkle Blooms Along the Kerala Coast

*Blooms of the bioluminescent plankton *Noctiluca scintillans*, locally called "sea sparkle" or kavaru, regularly light up the backwaters and beaches of Kochi and other parts of the Kerala coast in a glowing blue, especially after the monsoon.*

The same phenomenon has been recorded at Bangaram Island in Lakshadweep, Juhu Beach in Mumbai and Thiruvanniyur Beach in Chennai, though marine scientists note it can also signal nutrient overload in coastal waters, a warning sign alongside the spectacle.

Because scientists cannot easily observe bioluminescent animals without disturbing them

(bright lights can cause mobile animals to flee and may damage light-sensitive eyes), much about the exact purpose of specific light displays remains under active study. Researchers broadly group its uses into defence (startling or confusing a predator, or making an animal harder to see from below through a technique called counter-illumination), offence (luring or spotting prey), and communication or courtship (signalling to potential mates or rivals).

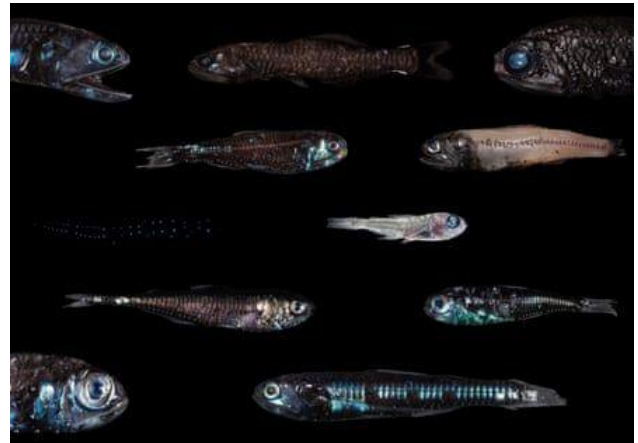


FIGURE 8: LANTERN FISH SPECIES

Species Spotlight

Lanternfish (Family Myctophidae): *Fuel for the Largest Migration on Earth*

Every night, lanternfish join trillions of other small deep-sea animals in swimming up toward the surface to feed, then retreat to darker water by dawn: a movement known as diel vertical migration, believed to be the single largest synchronised movement of animal biomass on the planet.

Scientists estimate there may be over 100 billion lanternfish in the world's oceans, making them one of the most abundant groups of vertebrates on Earth despite being largely invisible to humans.

Source: NOAA Ocean Exploration: What is vertical migration of zooplankton and why does it matter?

Deep-Sea Anglerfish: *A Fishing Rod Powered by Borrowed Bacteria*

The glowing lure dangling above an anglerfish's mouth, called the esca, does not produce light itself: the fish cannot generate bioluminescence on its own, and instead cultivates a colony of light-producing bacteria that live inside the lure.

In many species, a male anglerfish that finds a female bites into her body and gradually fuses with her permanently, becoming a small, parasitic source of sperm for the rest of his life, an extreme adaptation to finding a mate in the vast, dark deep sea.

Source: MBARI: Deep-sea anglerfish, Animals of the Deep

Quick Quiz

1. What percentage of deep mid-water animals are estimated to be bioluminescent?
2. What three ingredients are needed for the bioluminescence chemical reaction?
3. What is counter-illumination, and how might it help an animal avoid predators?
4. Why is bioluminescence difficult for scientists to study directly in the wild?

Key Terms

Bioluminescence: The production and emission of light by a living organism through a chemical reaction

Luciferin / luciferase: The light-emitting molecule (luciferin) and the enzyme (luciferase) that react together, with oxygen, to produce bioluminescent light

Photophore: A light-producing organ found in many deep-sea animals

Counter-illumination: Using bioluminescence on the underside of the body to match faint light from above, making an animal harder for predators below to see in silhouette

Things to Remember

1. Roughly 80 percent of animals living in the deep mid-water zone are bioluminescent.
2. Bioluminescence is produced by a chemical reaction between luciferin, oxygen and the enzyme luciferase.
3. Bioluminescence has evolved independently many dozens of times across very different marine groups.
4. Its likely uses include defence, hunting, and communication, though scientists are still studying many specific cases.

SECTION 5:

Regeneration

Can organisms regrow body parts?

Many marine animals can regenerate, or regrow, body parts they have lost, an adaptation that offers a powerful alternative to fighting off every threat directly. Sea stars are widely known for regrowing lost arms, but sea cucumbers take regeneration to a remarkable extreme. When threatened, a sea cucumber can eject some of its own internal organs, including parts of its intestine, respiratory structures and reproductive organs, through its body wall, either as a physical decoy that distracts a predator or, in some species, as a way to release sticky, toxic threads called Cuvierian tubules that entangle the attacker.

India Spotlight

Sea Cucumbers of the Gulf of Mannar

Sea cucumbers are so heavily harvested for the international bêche-de-mer trade that India banned all fishing and trade of them in 2001, placing them under Schedule I of the Wildlife Protection Act, 1972, the same level of legal protection given to tigers.

Before the ban, sea cucumber collection was a livelihood for roughly 200,000 fisherfolk in the Gulf of Mannar and Palk Bay region of Tamil Nadu; illegal poaching continues today, with single seizures in Lakshadweep worth over a crore of rupees.



FIGURE 9: SEA CUCUMBER

Losing internal organs would be fatal for most animals, but researchers studying sea cucumbers have found that they can regrow this lost tissue in as little as roughly six weeks, with the digestive tract regenerating from a small remnant left behind after evisceration. To understand how this is possible, scientists have sequenced the genomes of several sea cucumber species. One large genome-sequencing study of the Japanese sea cucumber, covering an estimated 92 percent of the species' full genome, found that sea cucumbers branched off from other echinoderms roughly 479 million years ago and carry notably fewer genes associated with hardening body tissue into mineral than their echinoderm relatives: a genetic difference researchers are still working to connect to the group's regenerative ability.

A separate, widely cited research programme has focused on *Holothuria glaberrima*, a species that regrows its intestine after evisceration and can also regenerate its radial nerve cord after injury. Researchers studying gene activity during this process have built detailed profiles of which genes switch on as new intestinal tissue forms, information that is now being compared with regeneration research in other animals, including some vertebrates, in the hope of better understanding why regenerative ability is so limited in humans and other mammals by comparison.

Species Spotlight

Common Octopus (*Octopus vulgaris*): Regrowing an Entire Arm, Nerves Included

A museum survey of preserved octopus specimens found that around 60 percent showed evidence of injury to one or more arms at some point in their lives, suggesting arm loss and regrowth is a routine part of an octopus's life rather than a rare event.

Octopuses can fully regenerate not just the muscle and skin of a lost arm but also its internal nerve cord, and researchers studying the process have found a specific enzyme that becomes unusually active in the regrowing

tissue, a discovery that may eventually inform human nerve-repair research.

Source: Scientific American: How Octopus Arms Regenerate With Ease

Common Starfish (*Asterias rubens*): *Choosing to Lose a Limb*

Starfish do not just regenerate arms lost to predators: researchers discovered in 2024 that a specific neurohormone actively triggers the starfish to voluntarily detach its own arm, a deliberate escape strategy called autotomy rather than an accident.

The tissue in the area where the arm detaches is thought to be biologically pre-adapted for rapid regrowth, as if the starfish's body is always prepared for the possibility of losing a limb.

Source: Popular Science: How starfish lose limbs (on purpose) and survive



FIGURE 10: COMMON STARFISH WITH A REGENERATED ARM.

Key Terms

Regeneration: The biological process of regrowing lost or damaged body parts or tissues

Evisceration: The expulsion of internal organs, used by some sea cucumbers as a defence mechanism

Echinoderm: A group of marine invertebrates that includes sea cucumbers, sea stars, sea urchins and sand dollars

Genome sequencing: A scientific technique used to read and map an organism's complete set of genetic instructions

Things to Remember

1. Sea cucumbers can eject internal organs as a defence mechanism and regrow them within weeks.
2. Genome studies have found sea cucumbers carry fewer hardening-related genes than related echinoderms.
3. Research on species such as *Holothuria glaberrima* focuses on which genes activate during regeneration.
4. Regeneration research in sea cucumbers has potential relevance to human regenerative medicine.

Quick Quiz

1. Why might a sea cucumber eject its own internal organs when threatened?
2. Roughly how long does it take a sea cucumber to regrow lost internal organs?
3. What group of marine invertebrates do sea cucumbers belong to, and name two relatives.
4. Why are scientists interested in studying sea cucumber regeneration in relation to human medicine?
5. What did the 2024 starfish study reveal about how autotomy is triggered?
6. What did the museum survey of octopus specimens suggest about how often arm injury occurs?

SECTION 6:

Communication

How do marine species communicate?

Underwater, sound is often a far more effective way to communicate than sight, because it travels roughly four times faster in water than in air and can carry across enormous distances, especially in the low frequencies used by large whales. According to NOAA Fisheries, marine mammals rely on sound as their primary way to communicate and to assess their surroundings, using it for finding mates, locating food, avoiding predators and staying in contact with other members of their group.

India Spotlight

Irrawaddy Dolphins of Chilika Lake

Chilika Lake in Odisha, Asia's largest brackish-water lagoon, holds the highest single-lagoon population of Irrawaddy dolphins in the world, monitored partly through underwater hydrophone surveys. Researchers who interviewed fishers across 30 villages and compared catch rates found that Irrawaddy dolphins appear to actively herd shoaling fish such as mullet toward stationary stake nets, a cooperative behaviour that benefits both the dolphins and the fishing community.



FIGURE 11: IRRAWADDY DOLPHINS OF CHILIKA LAKE

Different whale groups produce different types of sound. Mysticetes, or baleen whales (such as humpback, blue, fin and North Atlantic right whales), produce low-frequency pulses, moans and whoops that can travel very long distances. Odontocetes, or toothed whales, including orcas, belugas and sperm whales, produce higher-frequency whistles and clicks, and additionally use

echolocation: sending out a sound and interpreting the returning echo to judge the distance, size and even texture of nearby objects. Whales also communicate non-vocally, for example by slapping their tails or bodies against the surface of the water, a sound that can be detected hundreds of metres below the surface.

Perhaps the most famous whale sound is the humpback whale song: long, structured sequences of sounds that, unusually, are produced only by males. These songs were originally thought to be used mainly for attracting mates, but they have since also been recorded during feeding and migration, and scientists continue to debate their full range of functions. Whale calls can also travel astonishing distances: because of a phenomenon called the deep sound channel, in which temperature and pressure layers in the ocean bend and trap sound waves, low-frequency whale calls can in some cases be detected thousands of kilometres from where they were made.

Sound is not only useful to marine life: it is also increasingly disrupted by it. NOAA Fisheries has identified rising ocean noise from shipping, sonar, and industrial activity as a major and growing concern, noting that some baleen whales change their migratory path to avoid loud noise sources such as seismic surveys, and that increased background noise can make it harder for whales to hear and understand each other's calls.

Species Spotlight

Bottlenose Dolphin: The Only Animals Besides Humans Known to Use Names

Each bottlenose dolphin develops its own individual "signature whistle" within the first few months of life, which it uses throughout its life to identify itself to others, much like a name.

In controlled experiments, dolphins could still recognise which individual a signature whistle belonged to even after researchers digitally stripped away all of the voice-like qualities of the sound, showing that the identity information

is carried by the whistle's shape rather than by a recognisable "voice."

*Source: PNAS: Janik, Sayigh & Wells,
Signature whistle shape conveys identity
information to bottlenose dolphins*

Snapping Shrimp: A Two-Inch Animal Louder Than a Gunshot

Snapping shrimp produce one of the loudest sounds of any animal in the ocean, reaching around 183 to 210 decibels, not with their voice but by snapping an oversized claw shut fast enough to collapse a bubble in the water.

Colonies of snapping shrimp create a constant crackling background noise in shallow coastal waters worldwide, and researchers have found that this noise can interfere with sonar and underwater listening equipment.

*Source: NOAA Sanctuaries: Oh Snap! What
Tiny Shrimp Can Tell Us About Habitat Health*



FIGURE 12: SNAPPING SHRIMP

Key Terms

Echolocation: A biological sonar system used by toothed whales and dolphins to locate objects by interpreting returning sound echoes

Mysticete: A baleen whale, such as a humpback or blue whale, that filters food from water using baleen plates

Odontocete: A toothed whale, such as an orca, sperm whale or dolphin

Passive acoustic monitoring: A non-invasive research method using underwater microphones (hydrophones) to record and study marine animal sounds

Things to Remember

1. Sound travels faster and farther underwater than in air, making it a highly effective form of marine communication.
2. Baleen whales produce low-frequency calls, while toothed whales produce higher-frequency whistles and echolocation clicks.
3. Humpback whale songs are produced only by males and can change and spread between populations over time.
4. Human-generated ocean noise is a growing concern because it can interfere with marine mammal communication.
5. Dolphins identify each other using individual signature whistles, and snapping shrimp produce some of the loudest sounds in the ocean using a claw-triggered bubble collapse rather than a voice.

Quick Quiz

1. Why is sound often more effective than sight for underwater communication?
2. What is the difference between how baleen whales and toothed whales produce sound?
3. What is echolocation, and which whales use it?
4. How does human-generated noise affect whale communication, according to NOAA Fisheries?
5. How did researchers prove that dolphins recognise a signature whistle's shape rather than the caller's voice?

SECTION 7:

Social Structures

Why form groups?

Many fish species spend at least part of their lives in groups, and scientists distinguish between two related behaviours. Shoaling describes any loose gathering of fish, of the same or different species, that stay together for social reasons. Schooling is more specific: it describes a tightly coordinated group of fish of the same species and similar size, swimming at the same speed, in the same direction, in a highly synchronised way. Researchers estimate that roughly half of all fish species shoal at some point in their lives, while true schooling behaviour is seen in only about a quarter of species.

Predator avoidance: is one of the most widely supported explanations for schooling. A tightly packed, synchronised group can confuse a predator's senses: overlapping movement, sound and even electrical signals from many fish moving together can make it harder for a predator to track and target any single individual, a phenomenon researchers describe as a confusion effect. Being part of a large group also mathematically reduces any individual fish's own risk of being the one caught, an effect sometimes called the dilution effect.

India Spotlight

Oil Sardine Shoals of the Kerala Coast

*The Indian oil sardine (*Sardinella longiceps*) forms dense shoals along the Kerala and Karnataka coasts and has, on average, made up around 37 percent of Kerala's entire marine fish catch by weight over a ten-year study period. The species supports around 145,000 fishers working from 222 coastal villages in Kerala alone, and its shoaling patterns are closely tracked because the fishery's yearly success or failure has a direct economic impact on these communities.*



Schooling offers other advantages too. Groups of fish can locate food sources more efficiently, since many individuals scanning an area increase the chance of finding prey or spotting new feeding opportunities that other members can then exploit. Large schools gathering for spawning also improve reproductive success by increasing the chance of successful fertilisation. There is even evidence that swimming in a coordinated formation may offer hydrodynamic advantages, helping some fish conserve energy compared to swimming alone. Interestingly, schooling is not limited to prey species: researchers studying predatory fish such as jacks and barracudas on coral reefs have found that predators, too, sometimes school, possibly to learn about new food sources from one another or to improve their own hunting success against schooling prey.

Species Spotlight

Sardines and Anchovies: *Bait Balls That Overwhelm Predators*

When threatened, small schooling fish such as sardines can form a tightly packed, swirling sphere called a bait ball, sometimes only a few metres across but containing thousands of fish, which can persist for less than a minute before scattering.

Predators including tuna, sharks, dolphins and diving seabirds often attack a bait ball from multiple directions at once, and researchers studying these events have found that the

constantly shifting shape of the ball itself seems to make it harder for any single predator to lock onto one target.

Source: *Living Oceans Foundation: Predatory Fish Schooling on the Great Barrier Reef*

Scalloped Hammerhead Shark: A Predator That Schools for Reasons Still Unknown

Unlike most sharks, scalloped hammerheads gather in schools of up to several hundred individuals at specific seamounts and islands, including the Galápagos Marine Reserve, mostly during the day before dispersing to hunt alone at night.

Scientists have proposed several explanations, including using the school as a meeting point for mating or navigating by sensing magnetic anomalies in the seafloor, but researchers studying the behaviour say why they school in the first place is still not fully understood.

Source: *Galápagos Conservation Trust: Scalloped Hammerhead Shark*

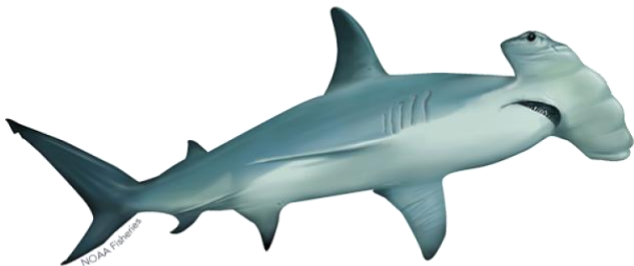


FIGURE 13: SCALLOPED HAMMERHEAD SHARK

Key Terms

Shoaling: A loose social gathering of fish, of the same or different species

Schooling: A highly coordinated, synchronised group of same-species fish swimming together

Confusion effect: The way a moving group can overwhelm a predator's senses, making it harder to target one individual

Dilution effect: The reduction in an individual's risk of predation simply by being part of a larger group

Things to Remember

1. Shoaling is a loose gathering; schooling is a tightly coordinated, synchronised group of one species.
2. Only about a quarter of fish species show true schooling behaviour.
3. Schooling can reduce predation risk through confusion and dilution effects, and can also improve foraging and reproduction.
4. Both prey and predator fish species are sometimes observed schooling.

Quick Quiz

1. What is the key difference between shoaling and schooling?
2. Explain the confusion effect and the dilution effect in your own words.
3. Name two benefits of schooling besides predator avoidance.
4. Why is scalloped hammerhead shark schooling considered unusual, and still not fully explained?

SECTION 8:

Migration

Why travel huge distances?

Migration is the regular, often seasonal movement of animals between different areas, usually driven by the need to find food, reach suitable breeding grounds, or avoid unfavourable conditions such as extreme cold. In the ocean, migration can cover astonishing distances: some sea turtles travel thousands of kilometres between feeding grounds and the very beach where they themselves hatched decades earlier, while some baleen whales migrate thousands of kilometres each year between polar feeding grounds and warmer breeding waters.

For many years, one of the great mysteries in marine biology concerned what scientists called the sea turtles' "lost years": the period after hatchlings leave their nesting beach and before they reappear as juveniles at coastal feeding grounds, sometimes years later. Researchers had long assumed that young Atlantic loggerhead turtles simply drifted passively within the North Atlantic Gyre, a large clockwise system of ocean currents. However, a NOAA-led study that attached small satellite tags to young turtles found that many of them actively left the gyre to spend time in the Sargasso Sea, a calm region within the gyre where drifting mats of sargassum seaweed offer shelter, food and, thanks to the seaweed absorbing the sun's energy, notably warmer water than the surrounding open ocean.

Migration timing is often closely tied to environmental cues. Many species move toward more nutrient-rich waters during spring and summer and return to warmer, more sheltered waters during autumn and winter. Because migration exposes animals to a wide range of conditions and threats across different regions and even different countries' waters, it also makes conservation more complex: protecting a migratory species in one part of its range does little good if it remains vulnerable elsewhere along its route.

India spotlight

Olive Ridley Arribada at Gahirmatha and Rushikulya, Odisha

The Gahirmatha Marine Sanctuary in Odisha is the world's largest rookery for sea turtles; in a single arribada (mass nesting event), several hundred thousand olive ridley turtles can come ashore over just a few nights.

The nearby Rushikulya rookery recorded its biggest arribada in 20 years of monitoring in 2025, with nearly 700,000 turtles nesting; scientists from the Zoological Survey of India have tagged around 12,000 turtles along the Odisha coast, and have already confirmed tagged individuals returning years later to nest again.



FIGURE 14: OLIVE RIDLEY SEA TURTLES COMING TO SHORE TO LAY EGGS - ARRIBADA AKA MASS-NESTING IN RUSHIKULIYA.



FIGURE 15: SATELLITE TAGGED FEMALE OLIVE RIDLEY TURTLE

Species Spotlight

Leatherback Sea Turtle: *The Longest-Distance Swimmer Among Sea Turtles*

One tagged female leatherback was tracked swimming more than 20,000 kilometres (about 12,700 miles) over 647 days, from her nesting beach in Papua, Indonesia, across the entire Pacific Ocean to feeding grounds off Oregon and back again, the longest migration ever recorded for a marine vertebrate at the time.

Leatherbacks feed almost entirely on soft-bodied prey such as jellyfish, and can dive to nearly 4,000 feet (about 1,200 metres), deeper than any other sea turtle.

Source: NOAA Fisheries: Leatherback Turtle Research

Gray Whale: *One of the Longest Migrations of Any Mammal*

Gray whales travel around 10,000 miles round-trip each year between Arctic feeding grounds and warm breeding lagoons in Baja California, Mexico, and some individuals have been tracked covering as much as 14,000 miles.

In 2013, a gray whale that normally lives in the North Pacific was spotted off the coast of Namibia in the South Atlantic, more than halfway around the world from where the species is normally found, in what researchers believe may be the longest single migration ever recorded for the species.



Source: NOAA Fisheries: Gray Whale

Key Terms

Migration: The regular, often seasonal movement of animals between different areas, usually linked to feeding, breeding or environmental conditions

North Atlantic Gyre: A large, clockwise system of ocean currents that circulates through the North Atlantic Ocean

Satellite tagging: A research technique that attaches a small tracking device to an animal to record its movements over time

Nesting beach: The specific beach where a sea turtle hatches and, as an adult, returns to lay its own eggs

Things to Remember

1. Migration is usually driven by the need to find food, reach breeding grounds, or avoid unfavourable conditions.
2. The sea turtle "lost years" refers to the poorly understood period between hatching and reappearance as a juvenile.
3. Satellite tagging revealed that young loggerhead turtles actively use the Sargasso Sea rather than merely drifting with the gyre.
4. Because migratory species cross many regions, protecting them requires cooperation across their entire range.

Quick Quiz

1. What environmental factors commonly drive marine migration?
2. What were the sea turtle "lost years," and how did NOAA researchers help solve this mystery?
3. Why does the Sargasso Sea offer young loggerhead turtles an advantage over the open gyre waters?
4. Why can migratory species be especially challenging to protect through conservation efforts?
5. How far did the tracked leatherback and gray whale travel, and why were both journeys considered records?

SECTION 9:

Navigation

How do species find their way?

Migrating across open ocean, with no landmarks and often no visible coastline, raises an obvious question: how do animals know where they are going? For sea turtles, decades of research point to one central answer: the ability to sense Earth's magnetic field, a sense scientists call magnetoreception. Earth's magnetic field varies in subtle, predictable ways from place to place, in both its intensity and its angle, or inclination, relative to the ground. Because each stretch of coastline has a slightly different combination of these two properties, researchers describe it as a kind of unique magnetic "signature" or "address" for that location.

Laboratory and field experiments have supported this idea. In one widely cited study, researchers glued small magnets to green sea turtles that had been displaced 100 kilometres from their nesting beach, disrupting their ability to sense magnetic fields; these turtles had significantly more difficulty finding their way home than turtles carrying a non-magnetic control weight of the same size. Scientists studying loggerhead turtles have proposed that hatchlings become magnetically imprinted on the field signature of their natal beach very early in life, and that as adults, they use this stored information to help guide them back to the same stretch of coastline to nest, sometimes called natal homing: even after journeys spanning thousands of kilometres and, in some cases, decades.

Indian Spotlight

Whale Shark Tagging Off the Gujarat Coast

Since 2011, the Wildlife Trust of India and the Gujarat Forest Department have satellite-tagged whale sharks rescued from fishing nets off Gujarat's Saurashtra coast, part of a long-running effort to understand the movements of the world's largest fish.

One female whale shark tagged near Sutrapada was tracked for over 200 days and roughly 5,500

kilometres, the longest whale shark migration ever recorded from the Indian subcontinent, travelling toward Somalia before turning back toward Gujarat.



Researchers caution that magnetic navigation, while well supported by evidence, is unlikely to be the only tool sea turtles rely on. Some studies have found that turtles carrying experimental magnets could still complete very large-scale migrations normally, suggesting that at a large scale, turtles may combine magnetic cues with other information, potentially including ocean currents and water chemistry, reserving especially precise magnetic navigation for the final stage of approaching a specific home beach. A separate large-scale analysis of nesting sites worldwide found that regions with more stable, more distinctive magnetic signatures over time tend to have more genetically distinct turtle populations, supporting the idea that magnetic navigation has shaped the population structure of these animals over many generations.

Species Spotlight

Sockeye Salmon: A GPS Written in Magnetic Memory

Researchers studying 56 years of fisheries data found that sockeye salmon returning to Canada's Fraser River choose between two possible routes around Vancouver Island, and that the proportion of fish taking each route shifted from year to year in step with small

natural drifts in Earth's magnetic field near the river mouth.

The leading explanation is that young salmon imprint on the magnetic field where they first enter the ocean, then seek out that same magnetic signature years later as adults, using it as a waypoint to guide them back toward their home river before switching to smell for the final approach.



Source: Oregon State University Newsroom: Salmon Use Earth's Magnetic Field to Navigate

Caribbean Spiny Lobster: *An Invertebrate With a Magnetic Map*

In a landmark experiment, researchers collected spiny lobsters, displaced them 12 to 37 kilometres to unfamiliar sites, and removed every cue they might use along the way, such as sight, smell and current direction; the lobsters still reliably oriented themselves back toward home.

This ability, called a magnetic map sense, makes the spiny lobster one of only a handful of invertebrates shown to be capable of true navigation, a feat otherwise documented mainly in vertebrates such as turtles, salmon and migratory birds.

Source: Nature: Boles & Lohmann, True navigation and magnetic maps in spiny lobsters

Magnetoreception: The ability of an animal to sense Earth's magnetic field

Magnetic imprinting: The process by which a young animal learns and stores the magnetic signature of its birthplace for use in navigation later in life

Natal homing: The tendency of an animal to return to the specific location where it was born, usually to reproduce

True navigation: The ability to determine one's position relative to a goal after being displaced to an unfamiliar location, without relying on familiar landmarks or cues from the destination

Things to Remember

1. Sea turtles can sense Earth's magnetic field, a sense known as magnetoreception.
2. Each coastal location has a distinctive magnetic signature based on field intensity and inclination.
3. Experiments using magnets attached to turtles have shown that disrupting magnetic sensing impairs their ability to navigate home.
4. Turtles likely combine magnetic cues with other information, relying most precisely on magnetic navigation near the final approach to their home beach.

Quick Quiz

1. What is magnetoreception, and why is it especially useful for open-ocean navigation?
2. What two properties of Earth's magnetic field give a coastline its unique "signature"?
3. Describe the magnet experiment used to test whether turtles rely on magnetic cues.
4. Why do scientists think turtles use more than just magnetic cues for navigation?
5. How did researchers rule out landmarks, smell and current cues when testing spiny lobster navigation?
6. How did the 56-year sockeye salmon study suggest fish use a magnetic imprint of their home river?

Key Terms